

STS EEG Screening Server Update

Global STD fix + image cascades — September 2026

Four steps on each server. Do the development server first; when its results look right, do production the same way. Everything is a file copy through your normal Remote Desktop session — no scripts required.

Each study is now processed in **two passes**: the brain panel is written first, in seconds, and the image cascade follows over the next few minutes. The panel reaches the practitioner straight away, and a cascade failure can no longer destroy it.

What this changes

- **Global STD** on page 1 of every Brain Panel currently prints 1000.00 with z 2642.71. After the update it reads about 2.5, inside the normal range. No other metric moves — verified 47 of 48 identical against this server's own output.
- **An image cascade** (<study>.imagecascade.pdf, one page per ICA component) is written next to every panel, a few minutes after the panel itself.
- **Disk**: roughly 10–15 MB per study. 1,000 studies use about 15 GB of production's 100 GB free. The new watchdog warns below 10 GB.

The files

On your workstation, open this folder in Explorer:

```
C:\BrainPanel\clean_eeg_project 2025\deploy\server-2026-09\staged
```

File	Goes into	What it does
files\edftotextbynameplotproc.py	project\files\	Global STD fix
files\edftotextbycommandplotproc.py	project\files\	panel pass only
files\Montage_6.py	project\files\	cascade rendering
files\dummy_gui.py	project\files\	cascade component pages
files\Component_selector.py	project\files\	same fixes, kept in sync
files\create_report_pdf.py	project\files\	cascade screen capture
run_cascade.py	project folder (next to module7.py)	new — the cascade pass
tomwatchdog_serialized.py	project folder (next to module7.py)	new watchdog — one at a time

“Project folder” is the directory containing module7.py. On the development server that is C:\app\MyCleanEEG\CleanEEGProject.

On the server

In your Remote Desktop session to the server.

1. Back up the six files being replaced

In the project's `files\` folder, make a new folder called `backup-2026-09` and copy these six files into it:

```
Component_selector.py
Montage_6.py
create_report_pdf.py
dummy_gui.py
edftotextbycommandplotproc.py
edftotextbynameplotproc.py
```

2. Paste the new files in

- Copy the six `.py` files from `staged\files\` into the project's `files\` folder. Say **Yes** to overwrite.
- Copy `run_cascade.py` and `tomwatchdog_serialized.py` into the project folder itself, next to `module7.py`.

3. Swap the watchdog

In the console window where `tomwatchdog.py` is running, press **Ctrl-C**. Then, in that same window:

```
set CLEANEEG_NO_BRAIN=1
```

That switches off the 3D source-localization pages. **The servers need it** — those pages render through VTK, which needs OpenGL 3.2+, and the AWS machines have no graphics card. VTK does not report an error; it kills the process. Do not set it on a workstation that has a real graphics card.

Now start the new watchdog. **Development server:**

```
py tomwatchdog_serialized.py "c:/app/STSEEGScreening/Source/Practitioners"
```

Production server:

```
py tomwatchdog_serialized.py
```

It prints `monitor dir:` and `interpreter:` first — glance at both. The production path is built in; the development server needs its own path given.

It must run in that console window — not as a service, not from a scheduled task. The cascade captures the screen, and only that session has one.

Once a study has processed cleanly, make the variable permanent from an elevated prompt: `setx CLEANEEG_NO_BRAIN 1 /M`.

4. Upload one study and check the result

The panel appears within seconds; the cascade follows. The console shows:

```
PROCESSING: ... (queue depth 0, N GB free)
```

```
PANEL in 6s rc=0
[CLEANEEG_NO_BRAIN set: skipping 3D source-localization pages]
CASCADE OK: ...imagecascade.pdf
  CASCADE in 66s rc=0
  DONE in 73s ...
```

Those timings are a short sample recording. A full-length study takes longer — the panel still in seconds, the cascade in minutes. Then check the files next to the EDF:

Dev	Prod	Check
☐	☐	.icale.rep.pdf — page 1, the Global STD row reads about 2.5 , not 1000.
☐	☐	.imagecascade.pdf exists, appearing a few minutes after the panel.
☐	☐	Open the cascade and page through it: component graphs on every page, no black pages .
☐	☐	Page 1 again: the four Moment 3 rows are in the hundreds to low thousands.

If the Moment 3 rows are in the **tens of thousands**: that server has numpy 2.x. Also paste staged-optional\process\detect_artifact.py into the project's process\ folder and re-run the study. One-time check per server.

That is the whole deployment. Production is the same four steps, starting the watchdog with no argument, in the console session kept logged in from the development server.

If something is wrong

1. Copy the six files from backup-2026-09 back into files\.
2. Ctrl-C the new watchdog.
3. Start the old one: py tomwatchdog.py

To keep the panel fix but stop cascades, set CASCADE_PASS = False near the top of tomwatchdog_serialized.py and restart it. Panels are unaffected.

Nothing in this update touches data files, the reference database, or any existing output, so rollback is a complete undo.

Worth knowing

- **Why two passes:** the cascade renders inside the same code that builds the panel, so in one pass the panel came last — and a cascade crash destroyed it, which is exactly what the OpenGL failure did on the dev server. Split in two, the panel is on disk before the cascade starts.
- **Why the watchdog changes:** the old one starts a new process for every upload with no limit. Two cascades rendering at once each capture the other's window, and the result is wrong with no error. The new one runs one study at a time, waits for an upload to finish writing, and kills anything past 25 minutes.

- **Development server only:** its screen is live only while you are connected. Stay connected while a cascade renders. Production is unaffected — its session is held by the console logged in from dev.
- **Screen saver / lock screen:** keep both off on any machine rendering cascades. A lock screen firing mid-render produces black pages.
- **Older panels:** every panel issued before this update carries the bogus Global STD row. Nothing else on them is wrong; their out-of-bounds count is one too high.
- **Longer documents:** README.md explains each change; MESA_OPENGL.md covers getting the 3D brain pages working on a server if you ever want them there.